

Facial Point Graphs for Stroke Identification

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Abstract. Stroke can cause significant damage to neurons, resulting in various sequelae that negatively impact the patient’s ability to perform essential daily activities such as chewing, swallowing, and verbal communication. Therefore, it is important for patients with such difficulties to undergo a treatment process and be monitored during its execution to assess the improvement of their health condition. The use of computerized tools and algorithms that can quickly and affordably detect such sequelae proves helpful in aiding the patient’s recovery. Due to the death of internal brain cells, a stroke often leads to facial paralysis, resulting in certain asymmetry between the two sides of the face. This paper focuses on analyzing this asymmetry using a deep learning method without relying on handcrafted calculations, introducing the Facial Point Graphs (FPG) model, a novel approach that excels in learning geometric information and effectively handling variations beyond the scope of manual calculations. FPG allows the model to effectively detect orofacial impairment caused by a stroke using video data. The experimental findings on the Toronto Neuroface dataset revealed the proposed approach surpassed state-of-the-art results, promising substantial advancements in this domain.

Keywords: Stroke · Facial paralysis · Deep learning · Facial Point Graph.

1 Introduction

Stroke is a neurological disease caused by an abnormality that changes the blood flow in the brain. The incident occurs when vessels that carry blood internally to this organ become clogged or rupture. There are two main types of strokes: ischemic and hemorrhagic. Ischemic strokes lead to insufficient blood supply in the affected area, resulting in a shortage of essential nutrients and oxygen. On the other hand, hemorrhagic strokes involve vessel rupture, causing increased pressure on the surrounding tissues and nerves. The disease is a matter of great concern worldwide and has escalated as the global population undergoes rapid aging [21]. Every year, there are approximately 10.3 to 16.9 million cases worldwide, with 5.9 million of them being fatal and requiring some form of treatment and therapy for 25.7 to 33 million cases [21]. In the US, the estimated annual cost is 34 billion dollars [2].

As a consequence of the disease, the cells in the affected brain region may incur damage, leading to various sequelae, including motor deficits, muscle atrophy, and facial paralysis [32]. These cells are crucial in coordinating many bodily functions, and their impairment can profoundly affect an individual’s overall health and well-being. Patients with facial paralysis experience weakened muscles that affect their ability to produce speech [24], significantly impacting an individual’s social communication abilities [30]. Moreover, everyday activities such as eating and drinking become more challenging, highlighting the importance of rehabilitation treatment for patients and creating automated processes to provide accessible treatment monitoring.

To address such a problem, researchers proposed automated image and video analysis techniques for evaluating orofacial impairments resulting from strokes and other diseases. For example, Bandini et al. [2] utilized three-dimensional facial landmarks extracted from video data and applied various feature calculations to assess facial symmetry and movement. Leveraging a Support Vector Machine (SVM) [5], they achieved an impressive 87% precision in classifying orofacial impairments in post-stroke patients. Similarly, Parra-Dominguez et al. [28] proposed a comparable approach involving calculating distances and angles related to facial symmetry. By extracting thirty features from patients’ images, they employed a MultiLayer Perceptron (MLP) [15] to detect facial paralysis, achieving an accuracy of over 94% on two different datasets. These advancements may improve treatment outcomes and enhance the overall quality of life for affected individuals.

The results obtained in the previous studies demonstrated impressive performance. However, it is important to consider that all the features employed were manually designed, which presents certain limitations. Handcrafted features heavily rely on human insights and may lack the adaptability required to handle real-world variations. In order to eliminate the need for such calculations, a model called Facial Point Graphs (FPGs) was proposed in [11] to cope with Amyotrophic Lateral Sclerosis (ALS).

The FPG is a model based on graphs formed from the most significant facial landmarks that use Graph Neural Networks (GNNs) [31] for inferring features from them. In this work, the model determines whether an individual has suffered a stroke or not, i.e., if there is orofacial impairment in the patient’s video while performing a specific task. To the best of our knowledge, there is currently no existing approach utilizing Facial Point Graph for detecting this impairment resulting from a stroke. Our strong conviction lies in the notion that the landmarks extracted from frames can be more effectively encoded within a non-Euclidean space, facilitating the accurate definition and representation of their unique characteristics. Hence, this paper has two main contributions:

- To introduce Facial Point Graphs for the identification of orofacial impairment arising from a stroke in videos.
- To employ a deep learning approach within this context, eliminating the need for manually crafted features.

The structure of this paper is as follows: Section 2 provides a comprehensive review of the relevant literature, and Section 3 offers a detailed explanation of the theoretical background. Besides that, Section 4 outlines the dataset used, describes the models employed for image cropping and facial feature extraction, presents our proposed approach, and explains the classification method. Section 5 shows the experimental results, while Section 6 encompasses discussing the results, drawing conclusions, and outlining avenues for future research.

2 Related Works

In the field of computer vision, the analysis of facial signals in images has aroused great interest and motivated various studies intending to automate the detection of facial paralysis [17, 28, 19]. Facial paralysis, caused by nerve damage, can arise due to congenital conditions, trauma, or diseases such as Bell’s palsy, brain tumor, or stroke. In this context, methods are developed considering facial asymmetry as a key indicator.

Parra-Dominguez et al. [28] proposed an MLP-based model to classify facial paralysis, disregarding the symmetric variations across different facial expressions performed by individuals. Their approach begins by extracting facial landmark features using a publicly available model called MEEshape [14], which accurately identifies 68 intrinsic points representing facial expressions in individual images. To account for potential head tilting in patient images, the authors incorporate a correction for the angle of the extracted facial points. Subsequently, calculations are performed, including measurements such as the distance from the corners of the eyes to the middle of the lower lip. Their MLP-based neural network achieves remarkable accuracy rates: 94.06% on the Massachusetts Eye and Ear Infirmary dataset (MEEI) [12], and 97.22% on the Toronto NeuroFace dataset [3].

Bandini et al. [2] conducted a study aimed at analyzing the impact of stroke on the orofacial musculature. The authors made significant progress in automatically evaluating this musculature. The study involved capturing videos of twelve stroke patients (seven males and five females) with an average age of 62.0 years and a standard deviation of 14.5. As a control group, videos of individuals without the disease were also included (seven males and four females) with an average age of 55.8 years and a standard deviation of 15.7. During the video recording sessions, participants were instructed to perform specific actions, including maintaining a resting position with normal biting and a neutral facial expression for twenty seconds (REST), repeatedly opening their jaw to its maximum five times (OPEN), puckering their lips as if kissing a baby for a total of 4 times (KISS), simulating blowing out a candle five times (BLOW), smiling with closed lips for five times (SPREAD), rapidly repeating the syllable “/pa/” in a single breath (PA), rapidly repeating the word “/pataka/” (PATAKA), and delivering ten repetitions of the sentence “Buy Bobby a puppy” in their usual tone and speaking speed.

Following a standardized procedure, the authors manually segmented each action repetition. For instance, the video of the SPREAD action was divided into five segments since the movement was executed five times. To ensure facial normalization and alignment of facial landmarks in each frame, the authors employed the Supervised Descent Method (SDM) algorithm [34], which aims to minimize the differences between the extracted features from each image. Ten geometric features were derived from the 3D coordinates of the facial landmarks for each repetition. Notably, many of these extracted features are related to facial symmetry, capturing potential signs of facial paralysis. Lastly, a validation method called “Leave One Out” (LOSO-CV) was employed with all 23 patients. This method involved treating one patient as the test set while utilizing the remaining patients as the training set for the machine learning model. The SVM model was employed in this study and achieved an accuracy of 87% in diagnosing the patients.

Pecundo et al. [29] conducted a similar approach. They classified the health status of individuals into three categories: ALS, Orofacial impairment in post-stroke, and healthy patients. The inference dataset used in their study was the Toronto Neuroface dataset [3]. The videos in this dataset showcased individuals performing various tasks, such as maximum jaw opening (OPEN), as already described above in this section. The authors manually segmented the videos since each contained multiple repetitions of the same subtask by an individual. Furthermore, they extracted specific facial landmarks using the Ensemble Regression Tree [18] implemented in the dlib library [20]. This model allowed the extraction of 68 facial landmarks, which were then used to calculate 14 essential features to infer ALS and Stroke and related to Facial Asymmetry, Range of Motion, and Mouth Shape and Geometry. The authors employed machine learning algorithms, including Random Forest [16], SVM, and K-Nearest Neighbors [10], achieving an accuracy of 86%.

As far as we know, the classification of stroke using a graph model formed by facial landmarks is innovative. However, GNNs models have already been used to capture emotional information. According to Ngoc et al. [26], facial landmarks formulated as graphs could result in an appropriate method for extracting valuable information from human expressions. In their model, the authors initially mapped the facial landmarks using Style Aggregated Network (SAN) [8]. It is worth noting that face landmarks formed the graph’s vertices; subsequently, the Delaunay triangulation method [6] was employed to establish the edges. Finally, a model called Directed Graph Neural Network was used for facial expression classification on the CK+ [23], MMI [27], and AFEW [7] datasets, achieving accuracies of 96.02%, 69.4%, and 32.64%, respectively. An important point to mention is that damage caused by a stroke to the facial nerve alters the facial expression since that nerve supplies the muscles of facial expression [22]. Given this knowledge, we employed a similar approach based on graphs constructed from facial points to analyze the impairments resulting from a stroke. This decision is based on validating similar methods in distinguishing facial expressions

related to emotions described above in this section. Such findings reinforce the GNN model as a promising approach for identifying orofacial impairments.

3 Theoretical Background

3.1 Graph Neural Networks

GNNs revolutionize how patterns are learned from a dataset by leveraging the power of graph representations. In the iterative process of GNNs, each node collects and integrates aggregated messages from its neighboring nodes using specialized aggregation and update functions. Importantly, each node transmits information to its adjacent nodes before updating its features. In the subsequent iteration, this new information (message) is again forwarded to its neighbors. This process of exchanging messages is visually demonstrated in Figure 1.

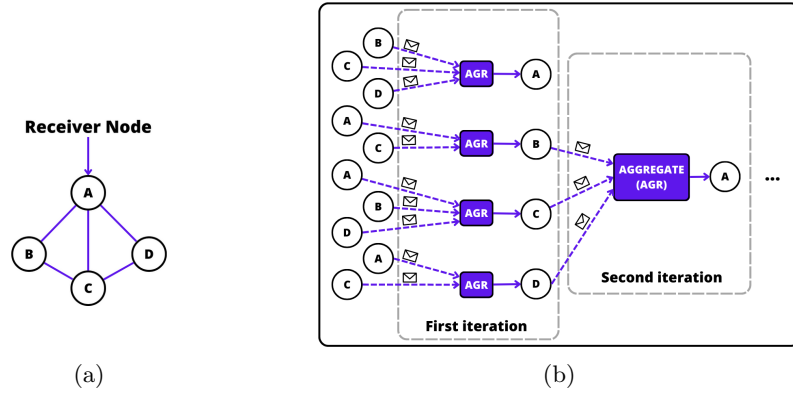


Fig. 1: Message aggregation in a bidirectional graph: (a) input graph and (b) working mechanism of a Graph Neural Network. For the sake of clarity, the second iteration focuses exclusively on node 'A'.

In each iteration k of the neural network, a hidden vector $\mathbf{h}_u^{(k)} \in \mathbb{R}^n$ is generated to capture the features of node $u \in \mathcal{V}$, where n represents the number of input features. Notably, the hidden vector $\mathbf{h}_u^{(0)}$ represents the initial encoding of the features before any training takes place. The process involves two steps: first, a node-order invariant function aggregates the features from the neighborhood $\mathcal{N}(u)$ of the node u , allowing the incorporation of relevant information from neighboring nodes. Secondly, the aggregated features are used to update the node's information, leading to the generation of $\mathbf{h}_u^{(k+1)}$. The specific mechanism for this update is described as follows:

$$\mathbf{h}_u^{k+1} = \text{Update}^k \left(\mathbf{h}_u^k, \text{Aggregate}_u^k \left(\{ \mathbf{h}_v^k, \forall v \in \mathcal{N}(u) \} \right) \right), \quad (1)$$

Equation 1 reveals the presence of both update and aggregation functions. While various models can be employed for the aggregation function, this paper adopts an attention-based formulation described further.

3.2 Graph Attention Networks

Graph Attention Networks (GATs) offer a powerful technique to enhance aggregation function by assigning different priorities to neighborhood information. This innovative concept was first introduced by Veličković et al. [33], who pioneered the development of a model with the following formulation:

$$\mathbf{h}_u^{k+1} = \sigma \left(\sum_{v \in \mathcal{N}(u)} \alpha_{v \rightarrow u}^k \mathbf{W}^k \mathbf{h}_v^k \right), \quad (2)$$

where $\mathbf{W} \in \mathbb{R}^{n' \times n}$ represents the weight matrix, which is a trainable parameter. Here, n' denotes the number of output features, and σ represents the sigmoid function. Additionally, $\alpha_{v \rightarrow u} \in \mathbb{R}$ denotes the attention given from node v to node u , indicating the degree of influence that v has on updating the features of u . A higher value of $\alpha_{v \rightarrow u}$ indicates a stronger impact of v on the feature update process of u . Mathematically, we can define $\alpha_{v \rightarrow u}$ as follows:

$$\alpha_{v \rightarrow u}^k = \frac{\exp \left(\text{LeakyReLU} \left([\mathbf{a}_u^k]^T [\mathbf{W}^k \mathbf{h}_u^k \parallel \mathbf{W}^k \mathbf{h}_v^k] \right) \right)}{\sum_{v' \in \mathcal{N}(u)} \exp \left(\text{LeakyReLU} \left([\mathbf{a}_u^k]^T [\mathbf{W}^k \mathbf{h}_u^k \parallel \mathbf{W}^k \mathbf{h}_{v'}^k] \right) \right)}, \quad (3)$$

where $\mathbf{a}_u \in \mathbb{R}^{2 \times n'}$ defines a trainable parameter known as the attention vector. The symbol $\parallel$ denotes the concatenation operator.

4 Experimental Methodology

4.1 Experimental Dataset

Introduced by Bandini et al. [3], Toronto Neuroface is a publicly available dataset designed to assess neurological disorders. The dataset comprises 261 colored videos, each clinically scored by two specialists. It includes oro-facial assessments, making it a valuable resource for tracking and analyzing facial movements in clinical populations affected by amyotrophic lateral sclerosis and post-stroke conditions. To validate the performance of our model, we use a balanced group comprising 11 individuals in the healthy control group (HC) and 11 individuals in the Stroke group. Each participant in the dataset performed multiple repetitions of speech and non-speech tasks. Important to note that not all participants in the dataset complete the tasks¹. After manually cropping the videos for each participant's task, we obtained 2,509 repetitions. The details of each task and the number of video repetitions are presented in Table 1.

¹ For the BLOW subtask, there were 15 participants, consisting of 8 stroke patients and 7 healthy controls (HC).

Table 1: Number of repetitions of each subtask.

Task	Subtask	Description	HC	Stroke
Speech	BBP	Repetitions of the sentence “Buy Bobby a Puppy”	111	104
	PA	Repetitions of the syllables /pa/ as fast as possible in a single breath	884	533
	PATAKA	Repetitions of the syllables /pataka/ as fast as possible in a single breath	275	163
Non-speech	SPREAD	Pretending to smile with tight lips	59	62
	KISS	Pretend to kiss a baby	57	62
	OPEN	Maximum opening of the jaw	55	61
	BLOW	Pretend to blow a candle	39	44

4.2 Pre-processing and feature extraction

In the pre-processing stage, we employ the OpenFace 2.0 tool [1] to filter out visual elements outside the subject’s face. This technique isolates the region of interest while incorporating head pose estimation to centralize the face. Consequently, the output of the pre-processing phase yields a grayscale cropped image, representing the bounding box region of the centralized main face, with dimensions of 200×200 pixels to each frame in the experimental dataset.

A series of feature extraction steps were essential to create the graph structure used in this study. Firstly, to capture landmark points, we employed Face Alignment Network (FAN) [4], a well-known deep learning approach for face alignment purposes. This method was built upon the hourglass (HG) network, designed initially for human pose estimation [25], using the landmark configuration of the MULTI-PIE 2D 68-point format [13] notation. Since the videos collected from the dataset only featured the frontal faces of individuals performing the tasks, we applied 2D coordinates landmark detection only.

For our study, as stroke facial paralysis can affect one or both sides of the face, we selected points on the face to capture the mirror positions of symmetry, comprising both sides of the forehead, eye, and mouth regions. Consequently, out of the 68 face landmark notations detected by FAN, we removed 20 landmark points irrelevant to our specific area of interest. This process resulted in a final set of 48 points, which we used as nodes in the graph. Each node was represented by its corresponding coordinates $[x, y]$, serving as their features.

The connections among nodes were established using the Delaunay triangulation technique [6]. This method generates a triangular mesh that effectively captures the spatial relationships among data points based on specific criteria. It relies on geometric considerations to determine whether a pair of neighboring triangles represents an optimal choice of connections, resulting in a favorable

representation of the spatial graph connection. In addition, a normalization technique was applied to facilitate the communication process among nodes during model training. The final structure of the graph ensures that all nodes were connected to point 31 of the MULTI-PIE notation (nose tip). This normalization point served as a reference to establish a consistent and standardized communication pattern among the nodes in the model. Finally, the Euclidean distance between each pair of nodes was calculated, serving as the weight for each edge in the graph. An overview of the entire process is presented in Figure 2.

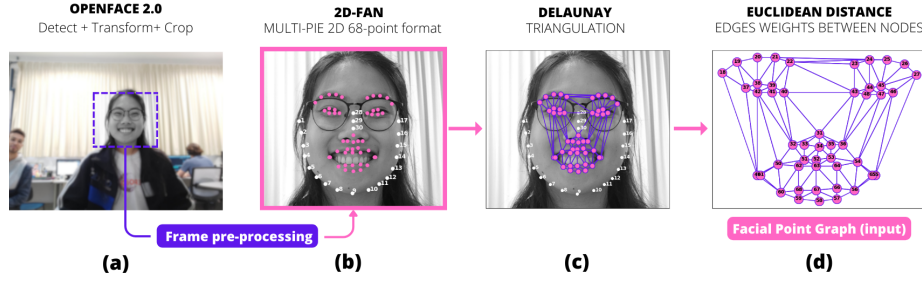


Fig. 2: Pre-processing and feature extraction procedures for constructing the graph structure used in the FPG model: (a) OpenFace 2.0 [1] detection, transformation, and crop of the main face. (b) Landmark extraction using 2D FAN [4] Facial Alignment method; (c) Delaunay triangulation [6] mesh process to connect landmark points of interest; and (d) Euclidean distance between each pair of nodes to calculate the edge weights of the final graph, used as input to the FPG model.

4.3 Classification and evaluation

As the classification was conducted on a video dataset, we employed the proposed approach by Bandini et al. [2] as our benchmark. Their approach utilized ten geometric features, four representing a range of motion (ROM) and six representing asymmetry, calculated in 3D dimensions. In our experiments, we employed handcrafted models for comparison purposes using the same feature extraction, but in 2D landmark instead due to our dataset limitation: SVM with linear and radial basis function (RBF) and Logistic Regression. For comparison purposes, we also applied the landmark coordinates features to train and evaluate the benchmark models. Moreover, a grid search was employed to find proper values for SVM parameters, i.e., the confidence value C and the RBF kernel scale parameter γ ².

The classification performance was evaluated using the LOSO-CV approach, ensuring separate splits for the training, validation, and test sets. This method-

² $C = [2^{-5}, 2^{-3}, \dots, 2^{15}]$, and $\gamma = [2^{-15}, 2^{-13}, \dots, 2^3]$.

ology effectively addresses concerns such as overfitting or memorizing training data. To maintain a balanced representation of both classes during the selection of the validation set, we randomly chose two subjects, one from the HC category and the other from the Stroke category, during each iteration split.

The evaluation was conducted in two forms, i.e., repetition- and subject-based classification:

1. **Repetition classification:** At each iteration of the LOSO-CV, the repetitions generated by one participant were regarded as individual samples in the test set. In this experiment, we calculated the hit/miss outcomes for each repetition (about each individual) to determine the final classification accuracy, ensuring that HC and Stroke participants and their respective repetitions were assessed in distinct test sets. Under this approach, speech and non-speech repetitions of individuals were classified as either belonging to the HC or Stroke group.
2. **Subject classification:** Following the same method proposed by Bandini et al. [2], we performed the LOSO-CV by treating each subject as a test case and classifying them based on the majority vote among the predicted repetitions, categorizing them as either HC or Stroke. In cases of ties, to ensure a more conservative prediction, the subject was considered HC.

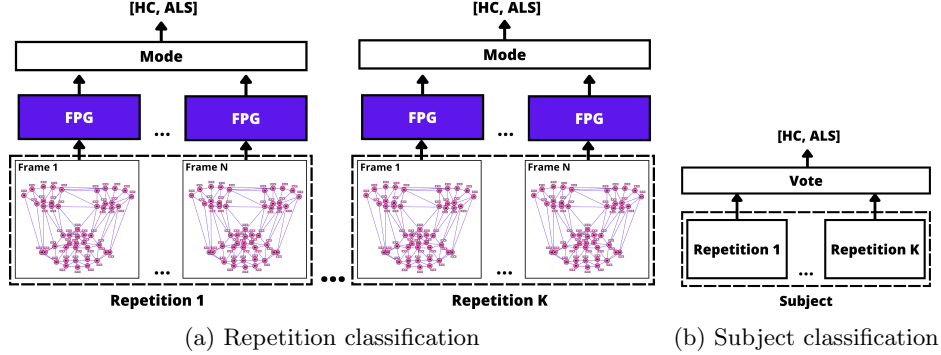


Fig. 3: Diagram representation of the two experiment modes.

In both repetition- and subject-based classification, we employed the validation set to mitigate bias in the model's hyperparameters and apply the early stopping technique. Additionally, we carefully selected the model's parameters based on empirical evaluation of its performance on the validation set. Therefore, the model is trained for a maximum of 50 epochs, using a batch size of 64 samples and setting the learning rates to 10^{-4} and 10^{-5} for the GAT and linear

layers, respectively. The hidden layers have a size of 32, and a dropout rate of 0.2 was applied after some of the LeakyRelu functions following the GAT layers.³

4.4 Proposed model

Initially, the proposed model uses fifteen equally spaced frames for each repetition performed by the patients. Therefore, each analyzed frame receives a graph of forty-eight nodes generated by the feature extraction step, as explained in Section 4.2. In this process, each of the fifteen graphs from frames of repetition video goes through six GAT layers and two linear layers. As one can notice in Figure 4, one crucial step before the information enters the linear layers involves pooling, where all feature vectors from the graph are combined into a single vector. This pooling operation is achieved through max pooling, which selects the highest value from each feature node and incorporates them into the pooled node vector.

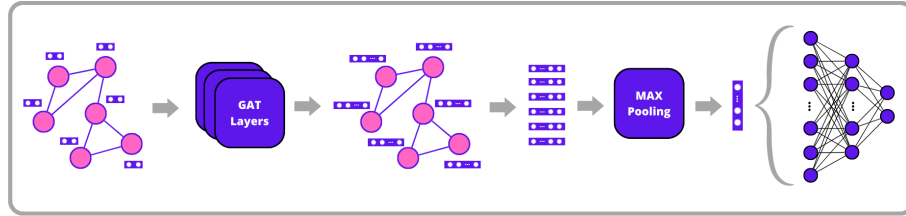


Fig. 4: FPG Model Approach

After pooling, the result moves through two additional linear layers, generating the model’s output. It is important to highlight that the error calculation is based on the class of each graph (frame). For repetition classification, the final results are obtained by identifying the mode among the frames used. This means that an individual’s repetition is classified based on the majority consensus among the frames, as depicted in Figure 3. Similarly, when classifying the subject, the majority mode is derived from the binary classifications of each repetition. This approach ensures a robust subject classification by considering the majority consensus across the various repetitions.

5 Results

The experimental results were obtained for each task, as shown in Table 2. It is worth noting that a standard number of frames used per repetition was estab-

³ Note: The experimental evaluation of FPG was conducted using PyTorch Geometric [9] on a GPU. An important point to mention is that, up to the current paper, the implementation of Graph Neural Networks (GNNs) using scatter operation on a GPU introduces non-deterministic behavior.

lished for comparing the baseline models with the FPG. Additionally, criteria such as accuracy, specificity, and sensitivity were considered to evaluate the results obtained by the proposed model.

Table 2: FPG results for each speech and non-speech tasks.

TASK		Classification	Accuracy	Specificity	Sensitivity
Speech	BBP	Repetition	55.3%	58.6%	51.9%
		Subject	59.1%	63.6%	54.5%
	PA	Repetition	69.3%	79.1%	53.1%
		Subject	81.8%	90.9%	72.7%
	PATAKA	Repetition	45.7%	55.3%	29.4%
		Subject	59.1%	63.6%	54.5%
Non-speech	SPREAD	Repetition	52.9%	55.9%	50.0%
		Subject	50.0%	54.5%	45.5%
	KISS	Repetition	64.7%	61.40%	67.7%
		Subject	72.73%	63.64%	81.82%
	OPEN	Repetition	58.6%	38.2%	77.0%
		Subject	63.6%	45.5%	81.8%
	BLOW	Repetition	65.1%	48.7%	79.5%
		Subject	73.3%	57.1%	87.5%

The proposed model achieved an accuracy of approximately 82%, with only three false negatives and one false positive for the ‘PA’ task, as presented in the confusion matrix in Figure 5. These findings indicate the development of a conservative model, meaning it is less likely to show the existence of facial paralysis when it does not exist. The binary classification task that stood out was the syllable ‘/pa/’ repetition. It is believed that the superior results in this classification are attributed to a larger quantity of distinct repetitions, as the patient pronounces the same syllable multiple times within a single breath. Deep neural networks, such as GNN, benefit from a larger volume of data, and in this specific case, the PA subtask has the highest number of repetitions compared to the others.

Other noteworthy results come from the BLOW and KISS tasks. The first task was also described by Bandini et al. [2] as one of the best tasks for predicting orofacial impairment caused by a stroke. By observing the confusion matrices presented in Figure 5, it can be seen that for these two tasks, the model indicated more correct classes for people who suffered a stroke, as also evidenced by the sensitivity of 87.5% for BLOW and 81.82% for KISS in Table 2. We firmly believe that, with the inclusion of more individuals in the healthy control group, the specificity will tend to increase, thus flourishing the accuracy of the model for these tasks.

One justification for the lower results in the other tasks is the classification of videos based on frames where there is not a sufficient extension of the orofacial muscles to infer orofacial impairment. Therefore, to validate the FPG, the

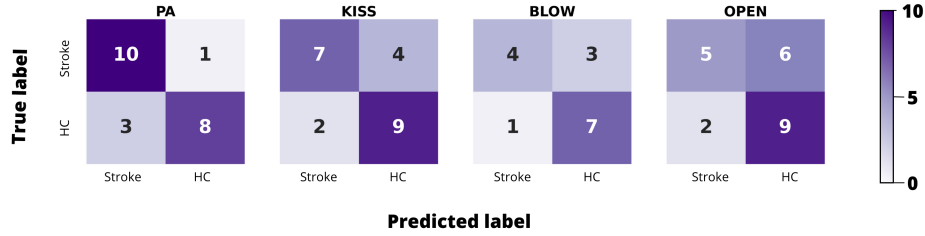


Fig. 5: Confusion matrices of subject-based classification for the most discriminative subtasks in our model.

experiments with SVM and Logistic Regression were conducted under the same conditions as the database used.

The SVM and Logistic Regression results are expected to differ significantly from the findings presented by Bandini et al. [2] due to a slightly different approach. Although Toronto Neuroface shares the same speech and non-speech tasks as Bandini et al.’s study [2], our method exhibits several distinctions. Firstly, the participants in our dataset were different. Additionally, we did not have access to videos containing three-dimensional depth features or samples from the REST subtask, which played a crucial role in normalization for both SVM and Regression models.

FPG achieved superior accuracy in comparison to the handcrafted feature extraction method proposed by Bandini et al. [2] across most classes. However, it’s noteworthy that FPG fell short of surpassing the performance of the SVM and Logistic Regression models in the OPEN and SPREAD subtask, as depicted in Figure 6a. An interesting insight emerges when we compare the outcomes of the benchmark models (Figure 6b), which rely solely on landmark coordinates as feature inputs. This comparison suggests that, for these specific subtasks, landmark coordinates alone may not offer sufficient discriminative information, clarifying the subtle performance distinctions between the benchmark and FPG modeling.

6 Discussion and Conclusion

This study represents the pioneering effort to evaluate GNNs for identifying Stroke impairments through video classification based on facial expressions. Using GNNs in feature extraction provides a pathway for reducing human bias and offers newfound possibilities for identifying stroke impairments in facial images.

The impact of stroke sequelae on daily activities, such as eating, drinking, and social interactions, can lead to social isolation and requires treatment to restore lost functions. In this pursuit, novel approaches like FPG emerged as promising means for identifying orofacial impairments caused by stroke, which is crucial for effective treatment. Adopting automated identification through

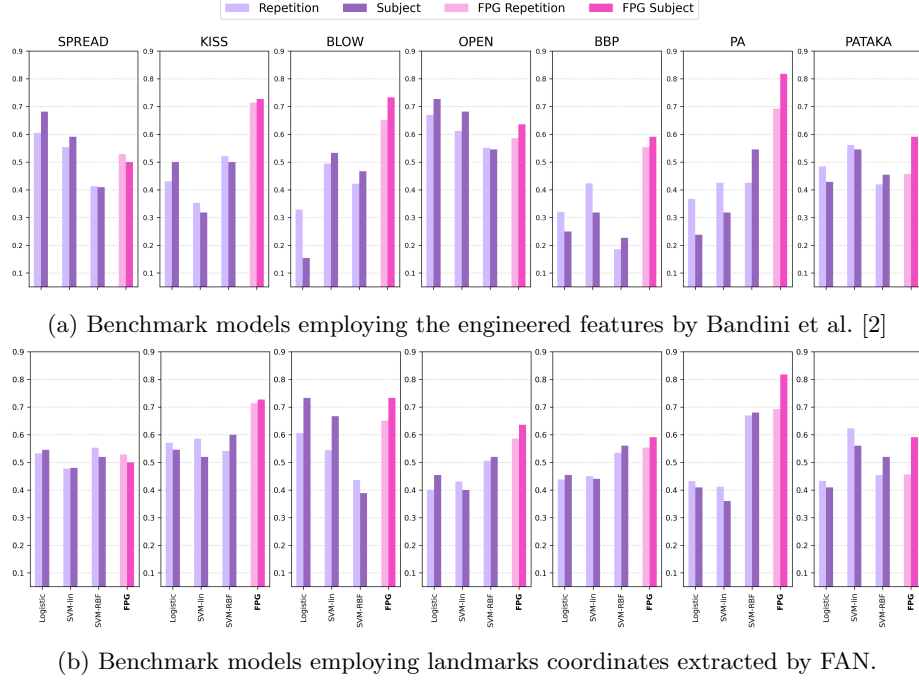


Fig. 6: Accuracy scores across the benchmark models and FPG model for each subtask.

FPG carries potential benefits, including improved patient care treatment and cost-effectiveness.

While Deep Neural Networks tend to struggle with generalization in smaller datasets, the graph-based FPG model achieved encouraging results, boasting an accuracy of approximately 82% in classifying orofacial impairments caused by stroke. Moreover, using two-dimensional data for feature extraction allows for more straightforward data collection using simpler cameras, broadening the scope for applications supported by readily accessible devices like smartphones.

In future works, it would be interesting to extend the analysis to other databases, aiming to infer facial impairment caused by paralysis originating from different diseases such as Bell's palsy or brain tumors. This expansion would contribute to enhancing the model's robustness. Another potential approach could involve analyzing only the frames where the most extensive movement occurs, specifically classifying the face image. This would reduce the model's dependence on frames where there is insufficient extension of the muscles affected by the disease, potentially improving the classification accuracy. These explorations could further refine and optimize the FPG model's performance, providing valuable insights for developing more accurate and clinically applicable stroke rehabilitation tools.

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